

Segment-Grass Effect: Anomalous Occupancy Sub-peaks at ± 0.2 rad

A Geometric Clipping Artefact in the Toy VELO Model

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Abstract

The hit-occupancy histograms produced by the `hit_competition_study` notebook exhibit small but persistent sub-peaks (“segment grass”) that appear *only* at the ± 0.2 rad angle setting and are absent at ± 0.1 and ± 0.04 rad. This note traces the effect to its root cause: **geometric clipping** of extreme-angle tracks on the outermost detector modules when the primary vertex is generated far upstream of the detector. We quantify the clipping probability analytically, confirm it with Monte Carlo, and show the direct track-level correspondence between clipped tracks and anomalous occupancy values.

1 Introduction

In the toy VELO tracking model, candidate segments are constructed between every pair of hits on adjacent detector modules. The *hit occupancy* — the number of candidate segments using a given hit — is a key diagnostic. For an event with n tracks per module, a hit on an internal module connects to hits on both neighbours, giving an expected occupancy of $\sim 2n$. Hits on the first or last module connect to one neighbour only, so their expected occupancy is n .

The `hit_competition_study` notebook revealed that at the ± 0.2 rad generation angle setting, the occupancy histograms contain small spikes at values below the expected n and $2n$ peaks. These “grass” features are completely absent at the narrower ± 0.1 and ± 0.04 rad settings. This note investigates the origin of these anomalies.

2 Geometry and Generation Parameters

The toy detector consists of $N_{\text{mod}} = 5$ planar modules at z -positions:

$$z_k = 100 + 33k \text{ mm}, \quad k = 0, 1, \dots, 4, \quad (1)$$

each with half-width $L_x = 50$ mm in both x and y .

Particles originate from a primary vertex (PV) whose coordinates are sampled from

$$z_{\text{PV}} \sim \mathcal{N}(0, \sigma_z^2), \quad \sigma_z = 50 \text{ mm}. \quad (2)$$

Crucially, this sampling is performed **independently for every generated event**. In the `hit_competition_study` notebook, the call

```
gen.generate_random_primary_vertices({"x": 0.1, "y": 0.1, "z": 50.0})
```

is made inside the per-event loop, so each event in the density sweep draws a fresh PV position. The argument `"z": 50.0` specifies $\sigma_z = 50$ mm (not a fixed PV position). As a result, the PV z -coordinate varies substantially across the ensemble: roughly 68% of events have $z_{\text{PV}} \in$

$[-50, +50]$ mm, but the tails extend far upstream. In particular, $\sim 38\%$ of events have $z_{\text{PV}} < -14.7$ mm — the critical threshold for module-4 clipping at ± 0.2 rad (see Section 3). This event-by-event variation in PV position is the direct cause of the variable number of clipped tracks and, consequently, the spread of anomalous occupancy values (“segment grass”) in the histograms.

Track angles (slopes) are sampled uniformly:

$$\phi \sim \mathcal{U}(-\phi_{\text{max}}, +\phi_{\text{max}}), \quad t_x = \tan \phi, \quad (\text{analogously for } \theta, t_y). \quad (3)$$

3 Root Cause: Geometric Clipping

A particle with slope $t_x = \tan \phi_{\text{max}}$ originating at z_{PV} reaches transverse position

$$x(z_k) = x_{\text{PV}} + t_x(z_k - z_{\text{PV}}) \quad (4)$$

at module k . The hit is recorded only if $|x| \leq L_x$. The maximum displacement occurs at the last module and at the largest angle. The critical PV z -position beyond which a track at $\phi = \phi_{\text{max}}$ misses module k is

$$z_{\text{crit}}^{(k)} = z_k - \frac{L_x}{\tan \phi_{\text{max}}} \quad (5)$$

For $\phi_{\text{max}} = 0.2$ rad ($\tan 0.2 \approx 0.203$), the critical values and associated clipping probabilities $P_{\text{clip}} = \Phi((z_{\text{crit}} - 0)/\sigma_z)$ are listed in Table 1.

Table 1: Critical PV z -positions and clipping probabilities for each module at $\phi_{\text{max}} = 0.2$ rad.

Module k	z_k (mm)	$z_{\text{crit}}^{(k)}$ (mm)	P_{clip} (%)
0	100	-146.7	0.2
1	133	-113.7	1.2
2	166	-80.7	5.3
3	199	-47.7	17.0
4	232	-14.7	38.5

For the narrower angle settings:

- $\phi_{\text{max}} = 0.1$ rad: $z_{\text{crit}}^{(4)} \approx -266$ mm ($> 5\sigma$, probability $< 10^{-5}\%$).
- $\phi_{\text{max}} = 0.04$ rad: $z_{\text{crit}}^{(4)} \approx -1019$ mm (effectively impossible).

This explains why the effect is observed *only* at ± 0.2 rad.

Figure 1 illustrates the PV z distribution overlaid with the module clipping thresholds.

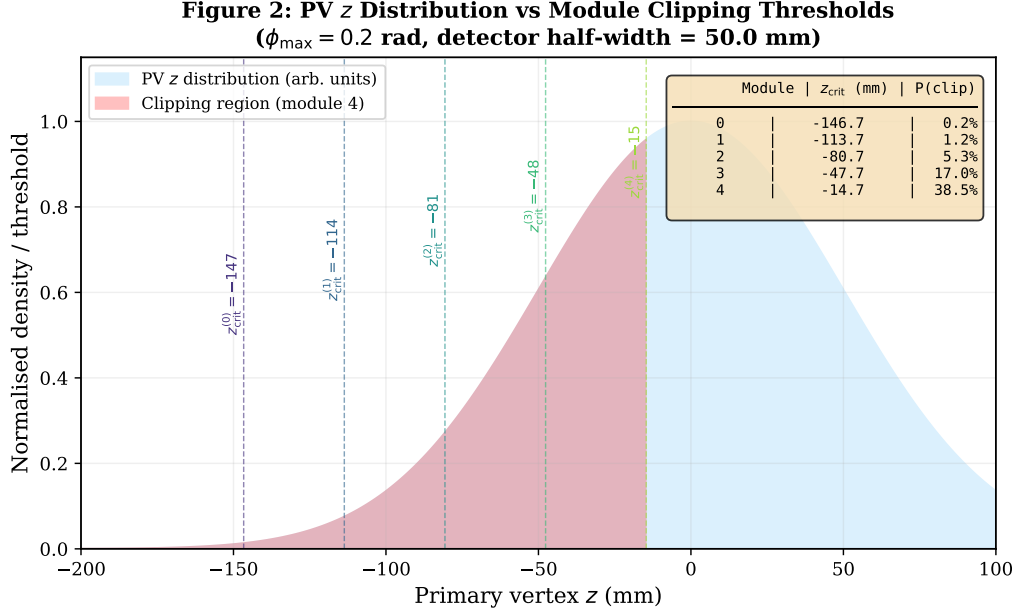


Figure 1: PV z distribution (blue shading) with clipping thresholds for each module (vertical dashed lines). The red-shaded region shows PV positions that cause at least one track to miss module 4.

4 Mechanism: How Clipping Creates Occupancy Sub-peaks

The segment construction creates all $n_i \times n_{i+1}$ segments between adjacent modules i and $i+1$, where n_i is the hit count on module i . For a hit on module k (internal), the occupancy is:

$$\text{occ}(k) = n_{k-1} + n_{k+1}. \quad (6)$$

When all tracks are accepted, $n_k = n_{\text{tracks}}$ for every module, giving the expected occupancy of $2n_{\text{tracks}}$.

However, when some extreme-angle tracks miss module 4, the hit count on that module drops: $n_4 < n_{\text{tracks}}$. This directly reduces the occupancy of every hit on module 3:

$$\text{occ}(3) = n_2 + n_4 = n_{\text{tracks}} + n_4 < 2n_{\text{tracks}}. \quad (7)$$

Since the number of clipped tracks varies event-by-event (depending on the PV z position and the specific random angles), n_4 takes a range of values across the ensemble, producing a spread of occupancy values below the expected peak. These are the “grass” sub-peaks in the histograms.

5 Direct Evidence

5.1 Event Displays

Figure 2 compares two events: one with the PV close to the detector (no clipping) and one with the PV far upstream (multiple clipped tracks shown in red).

Figure 1: Event Displays — Track Clipping at ± 0.2 rad

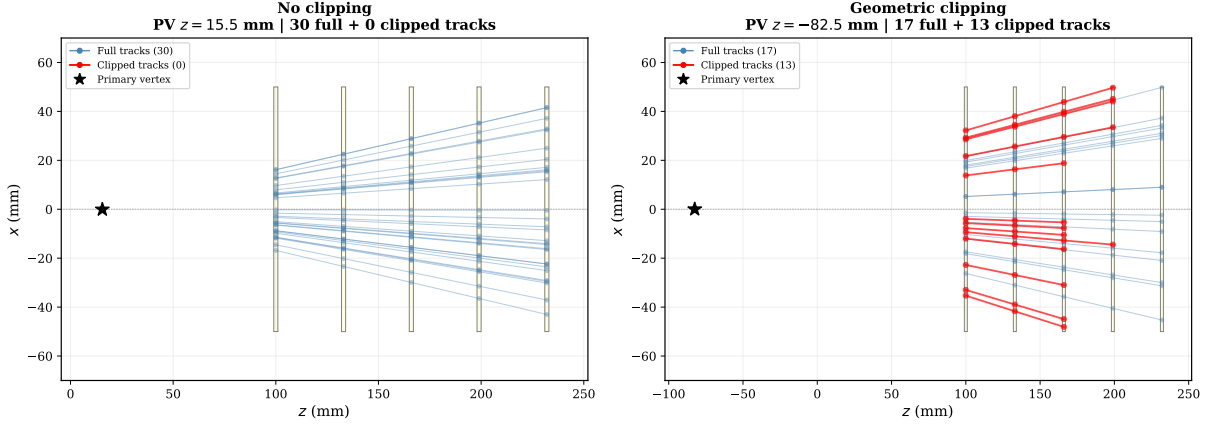


Figure 2: Event displays for a clean event (left) and a clipped event (right). Red tracks miss one or more modules. The gray dashed lines indicate the detector edge at ± 50 mm.

5.2 Track-Level Diagnosis

Figure 3 presents a complete diagnosis for a single event with clipping:

- Event display with clipped tracks (red) and their extrapolated miss positions ($\times$ markers beyond the detector edge).
- The (ϕ, θ) distribution of all tracks. Clipped tracks (triangles) cluster near the angular boundaries at $|\phi| \approx 0.2$ or $|\theta| \approx 0.2$.
- Box plot of per-module occupancy. Module 3's occupancy dips below the expected $2n$ because module 4 has fewer hits.
- Overall occupancy histogram for this event. Red bins mark the anomalous values arising from clipping; each one can be traced directly to the reduced n_4 count.

Figure 4: Track-Level Diagnosis — Connecting Clipped Tracks to Occupancy Anomalies

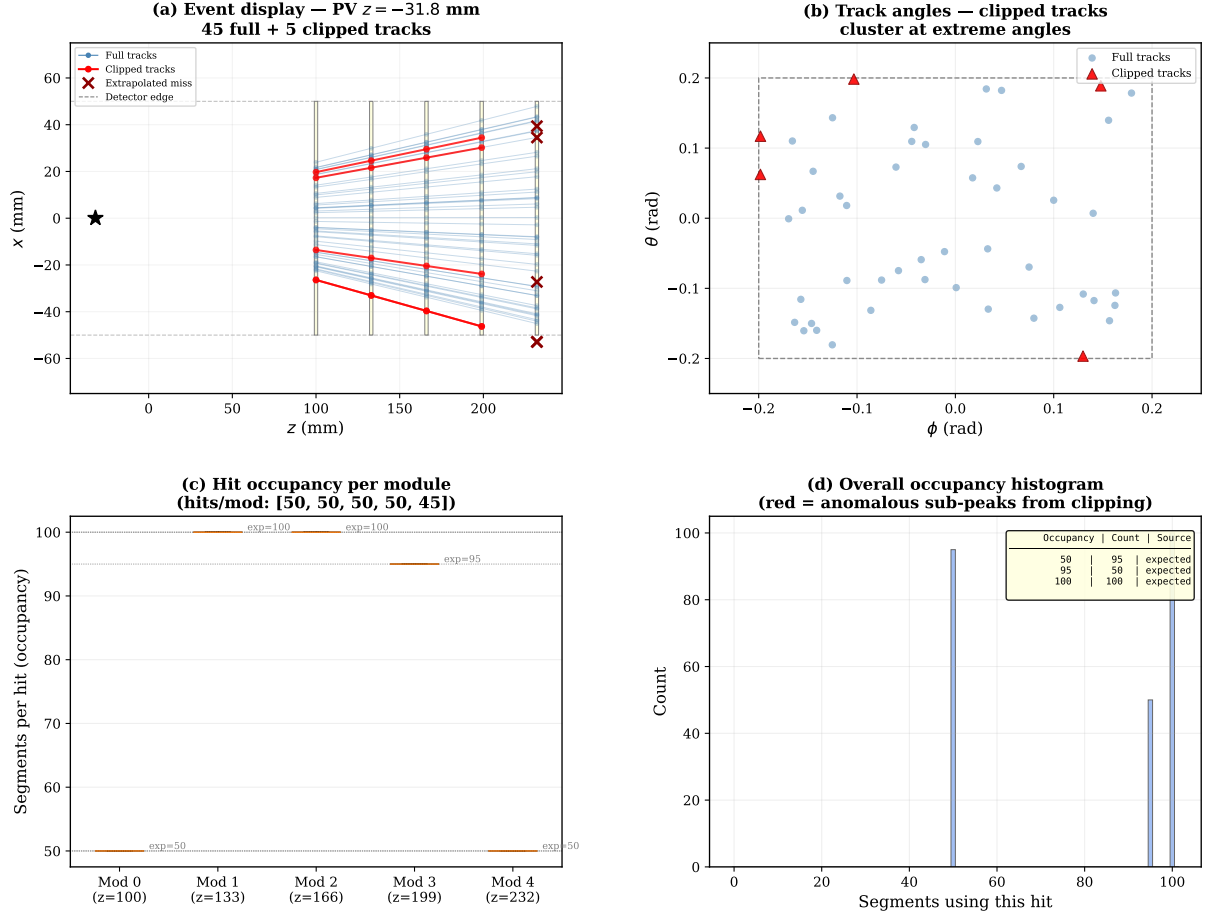


Figure 3: Complete track-level diagnosis. (a) Event display. (b) Track angles — clipped tracks (red triangles) lie near the ± 0.2 rad boundaries. (c) Per-module occupancy box plots. (d) Overall occupancy histogram with anomalous bins in red.

5.3 Occupancy vs Primary Vertex Position

Figure 4 shows the mean module-3 occupancy as a function of PV z across 200 events. The occupancy is constant at the expected value of $2n$ for $z_{\text{PV}} > z_{\text{crit}}^{(4)}$ and decreases monotonically for more negative PV positions, confirming the geometric origin.

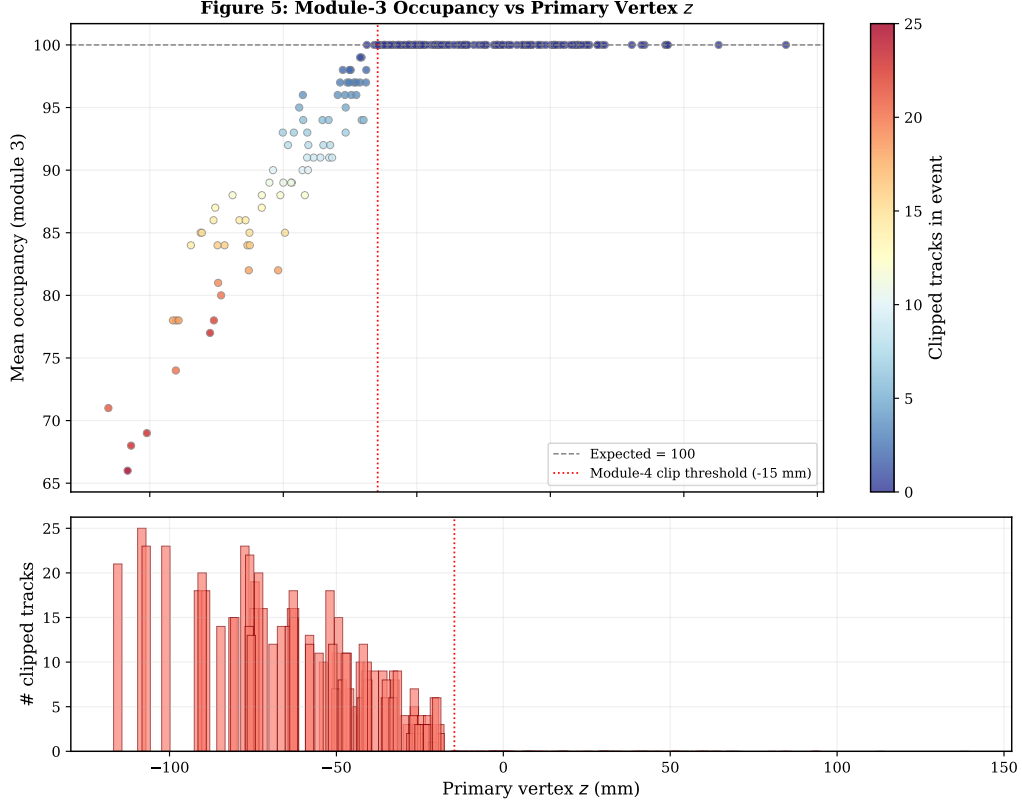


Figure 4: Module-3 mean occupancy (top) and number of clipped tracks (bottom) vs PV z position. Points are coloured by the number of clipped tracks. The red dotted line marks $z_{\text{crit}}^{(4)} = -14.7$ mm.

5.4 Angle-Setting Comparison

Figure 5 compares the occupancy histograms for all three angle settings (± 0.2 , ± 0.1 , ± 0.04 rad) with 50 tracks per event. Anomalous sub-peaks (red) appear only in the ± 0.2 rad setting.

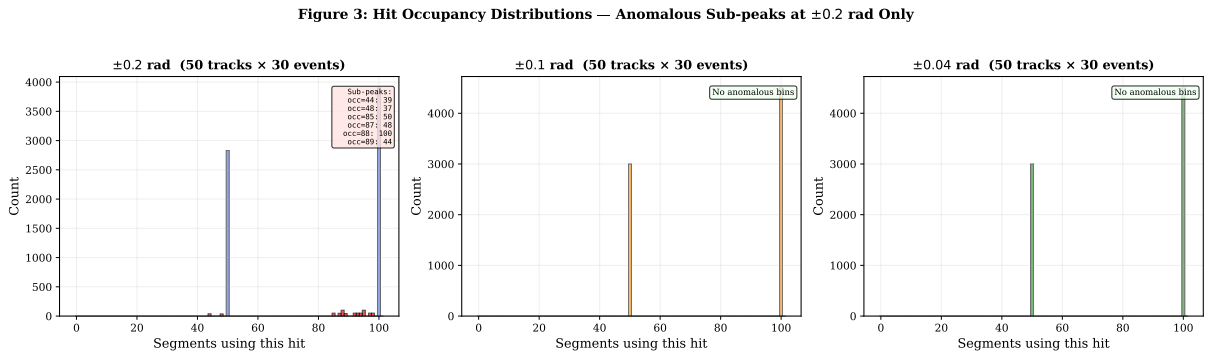


Figure 5: Occupancy histograms for three angle settings. Red bins indicate anomalous values absent from the expected peaks. Only the ± 0.2 rad setting exhibits sub-peaks.

5.5 Module Hit-Count Variability

Figure 6 confirms that the per-module hit counts are strictly constant at n_{tracks} for the narrower angle settings, but fluctuate on modules 3 and 4 at ± 0.2 rad.

Figure 6: Hits per Module — Variable at ± 0.2 rad (modules 3–4), Constant at Narrower Angles

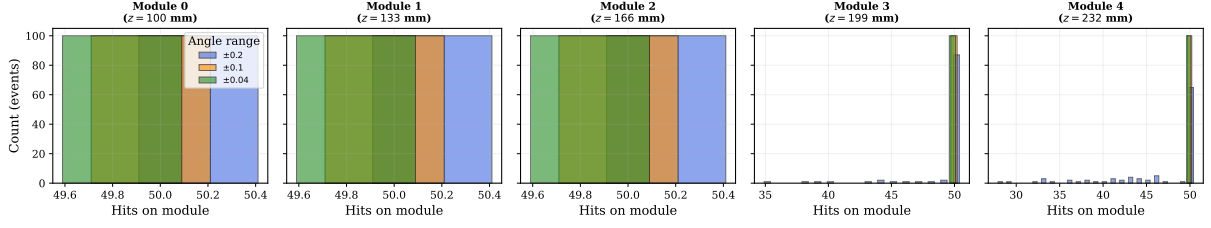


Figure 6: Distribution of hits per module across 100 events for each angle setting. At ± 0.2 rad (blue), modules 3 and 4 show variable hit counts due to geometric clipping.

6 Systematic Sweep: Ghost Rate and Drop Rate

To understand how measurement noise interacts with the geometric clipping effect, we perform a systematic sweep over two noise parameters: *ghost rate* (fraction of spurious hits added uniformly within module bounds) and *drop rate* (fraction of true hits randomly removed). The sweep covers a 6×6 grid: ghost rate $\in \{0, 5, 10, 15, 20, 30\}\%$ and drop rate $\in \{0, 5, 10, 15, 20, 30\}\%$, at two angle settings (± 0.2 and ± 0.1 rad) and three track multiplicities ($n = 20, 50, 100$), giving 216 independent parameter points. Each point averages over 10 events. The sweep was executed on HTCondor.

6.1 Occupancy Histograms Under Noise

Figure 7 reproduces the original hit-competition histograms from the clean baseline (ghost = 0, drop = 0), showing the segment-grass sub-peaks at ± 0.2 rad and their absence at ± 0.1 rad. Figure 8 overlays several noisy configurations on the same axes, demonstrating how ghost hits broaden the distribution (shifting counts rightward) while dropped hits erode the dominant peaks.

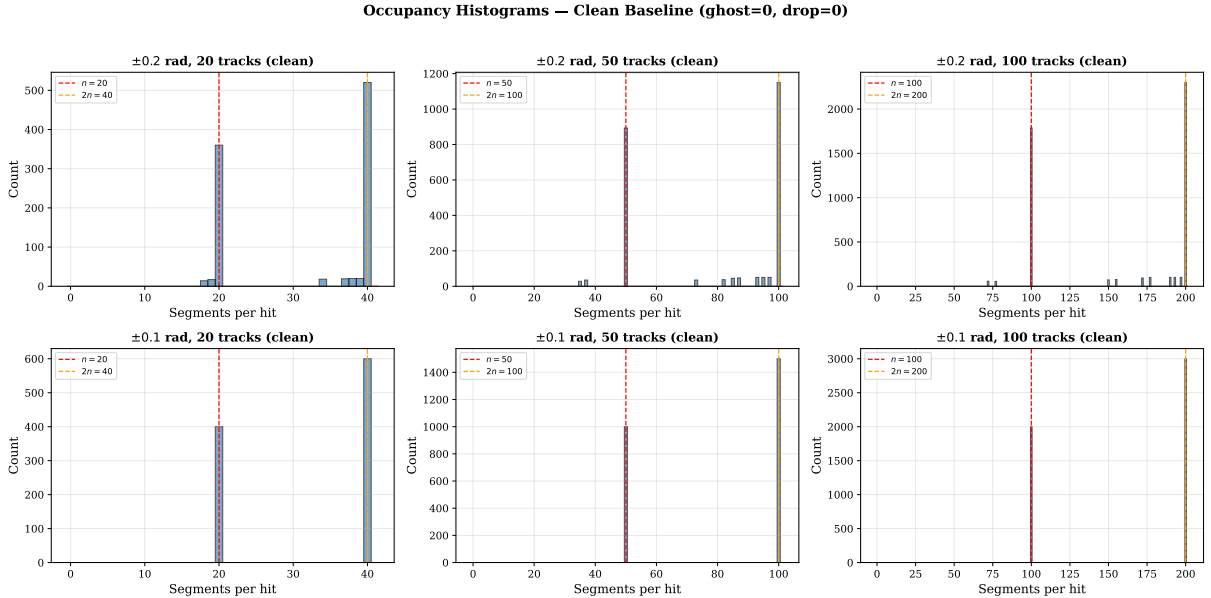


Figure 7: Clean-baseline occupancy histograms (2×3 grid: rows = angle settings, columns = track counts). Dashed lines mark the expected n and $2n$ peaks. Sub-peaks visible at ± 0.2 rad (top row) are the segment-grass artefact.

Occupancy Distributions — ± 0.2 rad: Clean vs Noisy

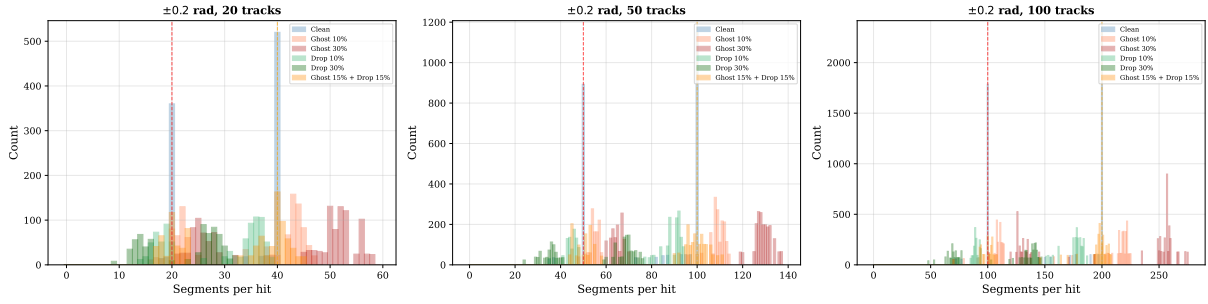


Figure 8: Occupancy distributions for ± 0.2 rad under various noise conditions. Ghost hits shift mass rightward (more segments per hit); dropped hits reduce the peak heights. The clean distribution (blue) is overlaid for reference.

6.2 Anomalous Occupancy Fraction

Figure 9 shows the fraction of hits whose occupancy falls outside the expected $\{n, 2n\}$ peaks. At the clean baseline, the ± 0.2 rad setting already has $\sim 4\%$ anomalous hits (from geometric clipping); any non-zero ghost or drop rate drives this above 70%. The ± 0.1 rad control has 0% anomalous hits when clean, confirming that all anomalous counts originate from noise injection rather than clipping.

Anomalous Occupancy Fraction vs Ghost Rate & Drop Rate

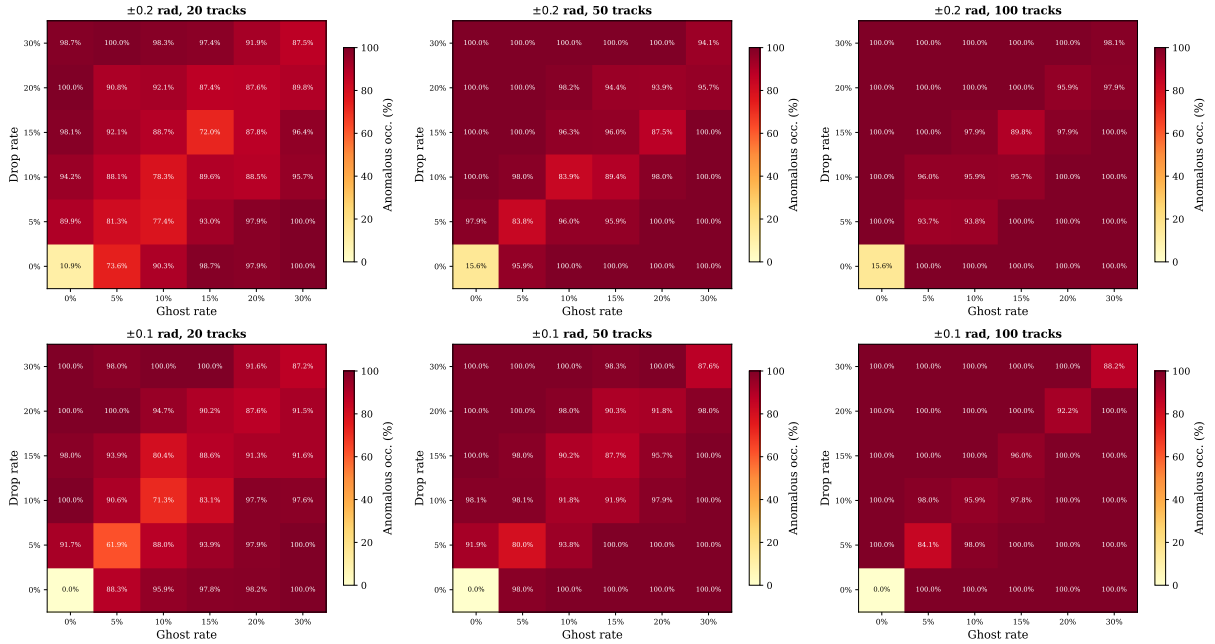


Figure 9: Anomalous occupancy fraction as a function of ghost rate (columns) and drop rate (rows). Upper panels: ± 0.2 rad; lower panels: ± 0.1 rad. Columns within each row correspond to increasing track multiplicity.

6.3 Reconstruction Efficiency

Figure 10 maps reconstruction efficiency across the sweep grid. Efficiency is robust to ghost hits (flat across ghost-rate columns at fixed drop rate) but degrades sharply with the drop rate: at 30% drop, only $\sim 30\%$ of true tracks are recovered regardless of ghost level. Notably, the

reconstructed ghost rate and clone fraction remain near zero across the entire sweep, indicating that the Hamiltonian solver does not produce spurious tracks even under heavy noise.

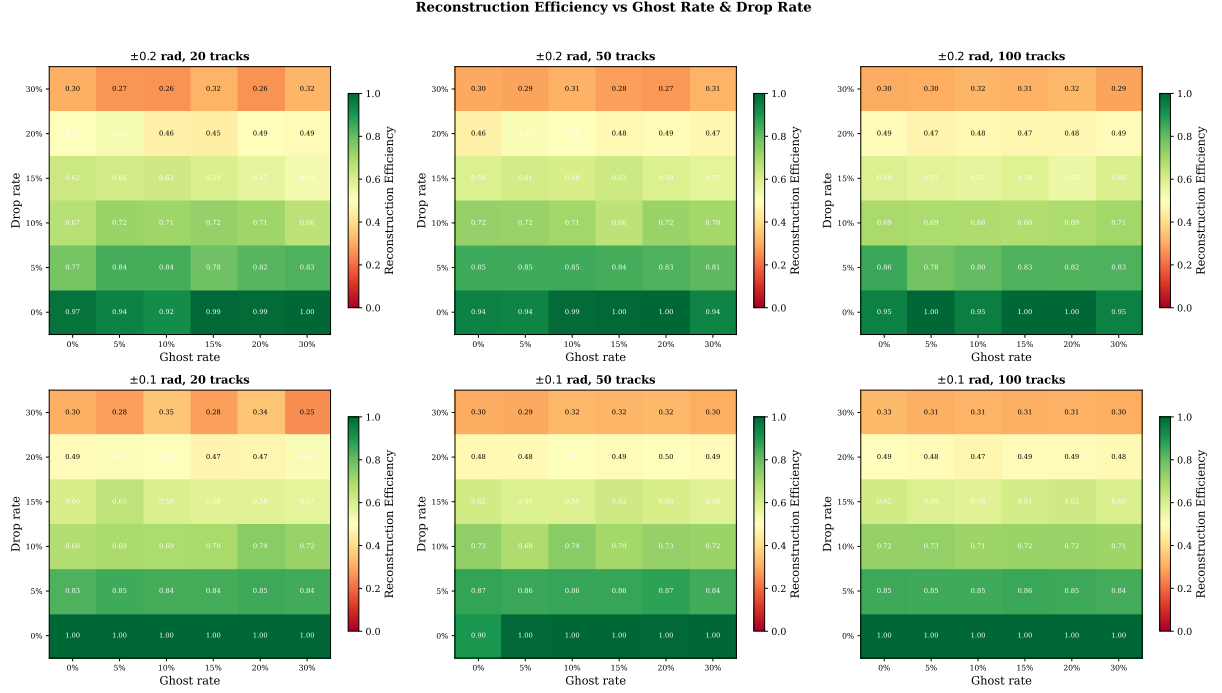


Figure 10: Reconstruction efficiency vs ghost rate and drop rate. Efficiency degrades primarily along the drop-rate axis. Ghost hits have minimal impact because the γ -term suppresses false segments effectively.

6.4 Segment Count Growth

Ghost hits inflate the Hamiltonian dimension. Figure 11 shows the mean number of segments. At 100 tracks with 30% ghost rate, the segment count reaches $\sim 65,000$ (compared to $\sim 40,000$ clean), a 60% increase. Dropped hits reduce segment count along the other axis, but the net effect of combined noise is dominated by ghosts.

Mean Number of Segments vs Ghost Rate & Drop Rate

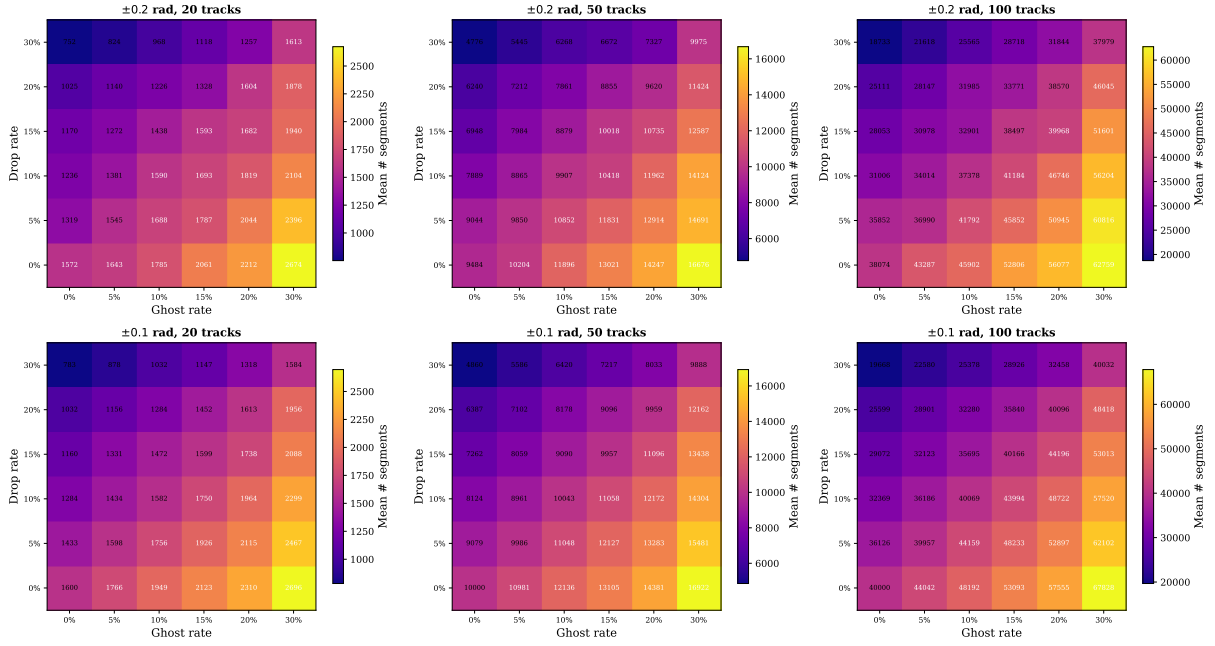


Figure 11: Mean number of Hamiltonian segments vs ghost and drop rate. Ghost hits create additional segments with all hits in adjacent modules, leading to super-linear growth.

6.5 Activation Spectrum

Figure 12 shows the mean activation of true and false segments as a function of noise. True-segment activations remain near 1.09 regardless of ghost rate, but degrade to ~ 0.83 at 30% drop rate as the loss of supporting hits weakens the self-interaction term. False-segment activations remain pinned at the baseline value $\delta/(\delta + \gamma) \approx 0.4$ across all noise levels, confirming that the competition mechanism suppresses false segments robustly.

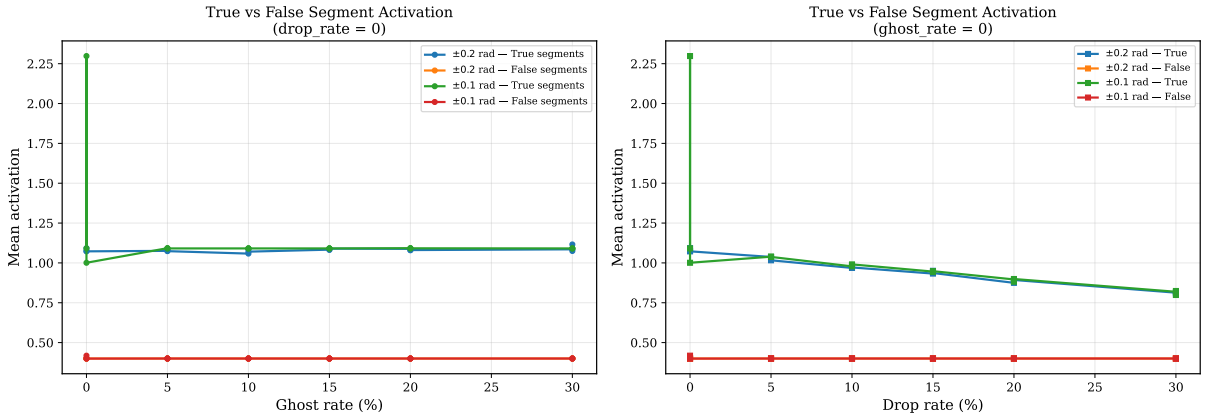


Figure 12: Mean activation of true (upper curves) and false (lower curves) segments. Left: varied ghost rate at zero drop. Right: varied drop rate at zero ghost. The gap between true and false narrows only with increasing drop rate.

7 Selected-Segment Corpus: Post-Solver Analysis

The preceding analysis (Sections 6 and earlier) characterises the *full* segment corpus produced by the Hamiltonian construction, i.e. every kinematically admissible doublet between consecutive

modules. Track reconstruction, however, is performed on the much smaller **selected corpus** of segments whose classical-solver activation x_i exceeds the threshold $T = 0.7$. This section repeats the previous diagnostics on this selected corpus, where the segment grass should — by construction — disappear.

The geometric expectation is sharp. In a 5-module detector, every hit on an internal module (M_1, M_2, M_3) participates in two surviving segments (one upstream, one downstream), and every hit on a boundary module (M_0 or M_4) participates in one surviving segment. The expected mean occupancy is therefore

$$\langle \text{occ}_{\text{sel}} \rangle = \frac{2 \cdot 1 + 3 \cdot 2}{5} = 1.6, \quad \text{occ}_{\text{sel}} \in \{1, 2\}.$$

Anomalous occupancy is now defined as $\text{occ}_{\text{sel}} \notin \{1, 2\}$.

7.1 Selected-Corpus Occupancy Histograms

Figure 13 shows the clean baseline. The distributions are bimodal at exactly the expected values, with mean ≈ 1.6 and standard deviation ≈ 0.5 across all configurations — an independent confirmation of the geometric prediction.

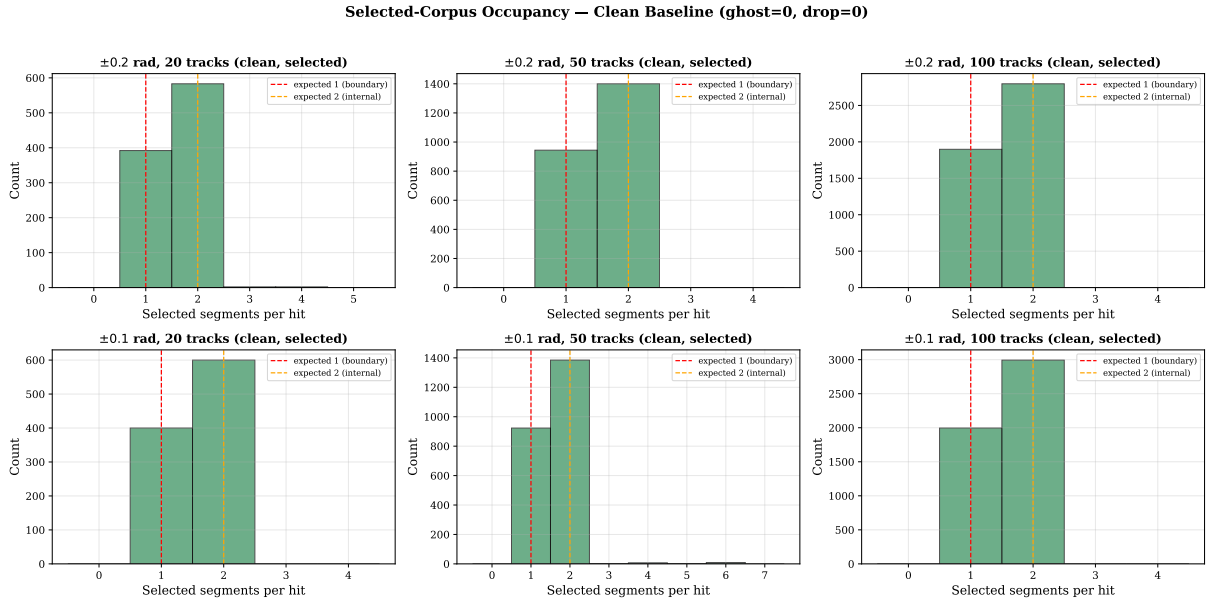


Figure 13: Occupancy histograms of the *selected* corpus ($x_i > 0.7$) at zero ghost and drop rates. All distributions concentrate on $\{1, 2\}$ as predicted by the geometry; no grass is visible.

Under noise (Fig. 14), the bimodal structure is preserved in shape but mass shifts toward $\text{occ} = 1$ as drop rate increases (fewer surviving segments per hit). Ghost-only noise barely perturbs the distribution at all.

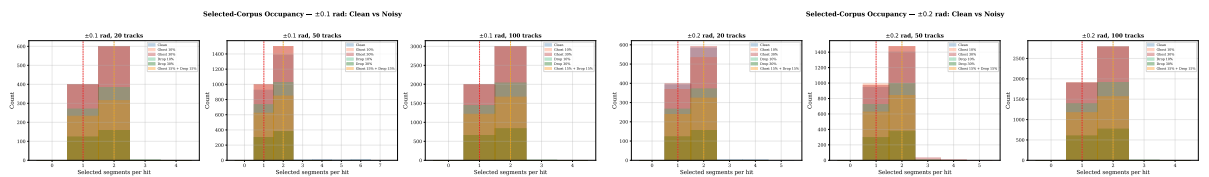


Figure 14: Selected-corpus occupancy across noise configurations at ± 0.1 rad (left) and ± 0.2 rad (right). The $\{1, 2\}$ structure survives all noise levels.

7.2 Anomalous-Occupancy Suppression

Figure 15 compares the fraction of hits with anomalous occupancy in the full and selected corpora. In the full corpus this fraction sits near unity across the entire grid (median 0.98); in the selected corpus it collapses to zero (median 0.0, mean 0.001). The classical solver therefore acts as a near-perfect grass filter: every selected segment's hits sit on $\{1, 2\}$ counts, just as the geometry predicts.

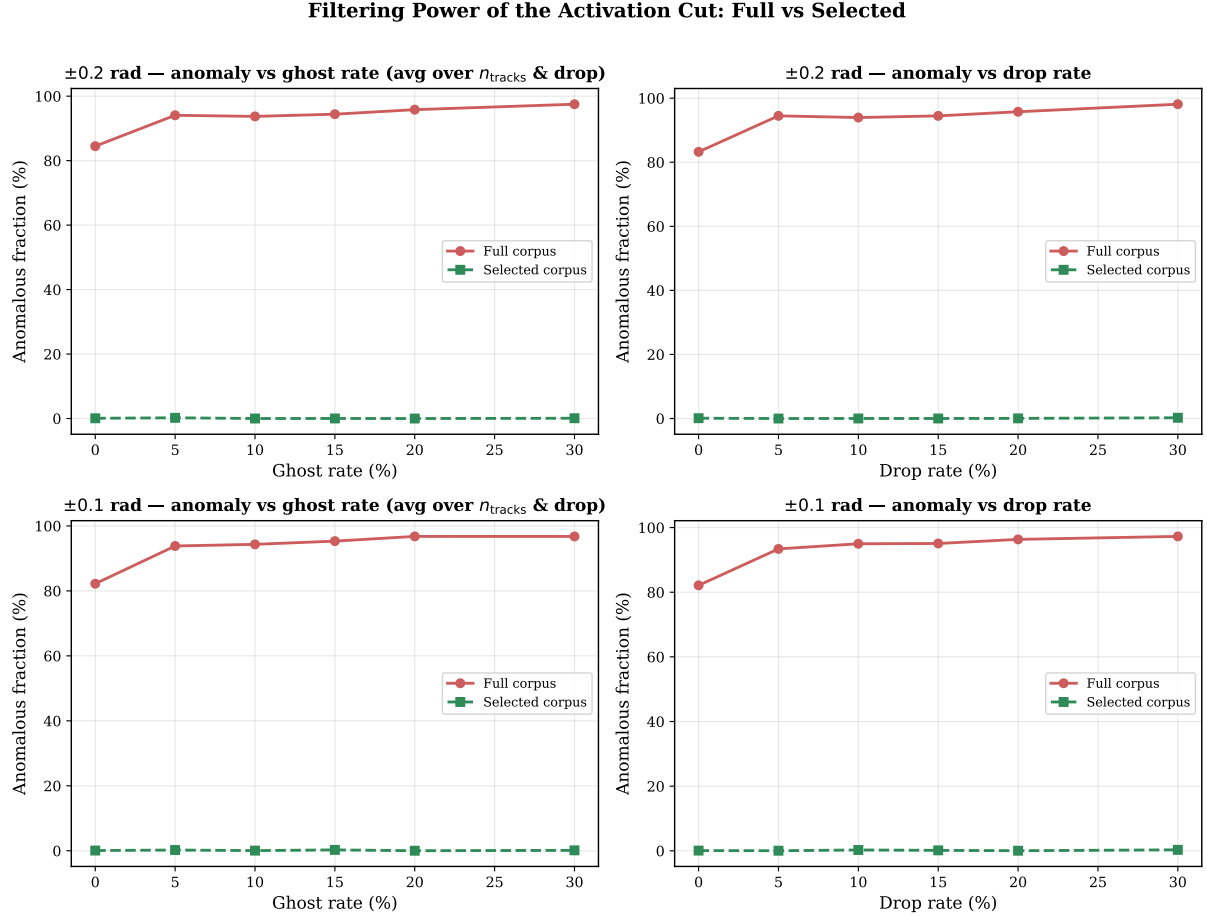


Figure 15: Fraction of hits with anomalous occupancy, full corpus (left) vs selected corpus (right), as a function of ghost and drop rates. Note the colour scales differ by two orders of magnitude.

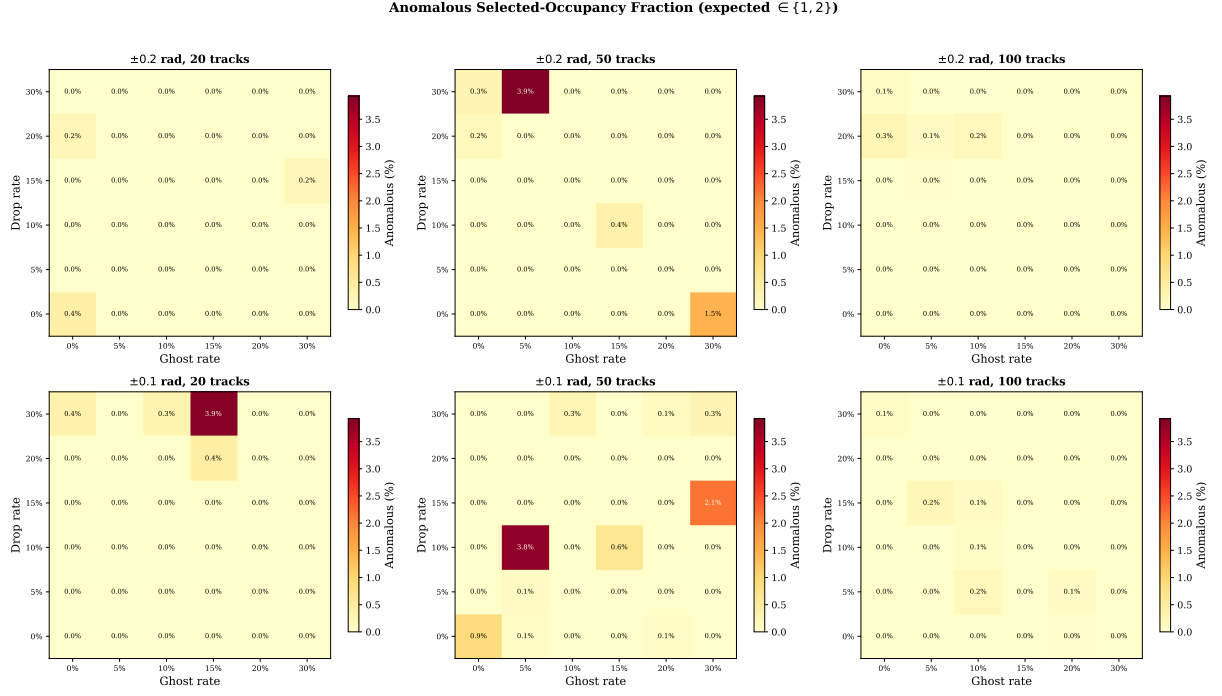


Figure 16: Selected-corpus anomalous-occupancy fraction across the full noise grid. Values are at or near the floor everywhere; the only structured residual is at the angle/ n_{tracks} boundaries where the solver occasionally promotes one extra false segment.

7.3 Selected Segment Counts and True/False Breakdown

Figure 17 shows the mean number of selected segments per event. In the clean baseline this matches $4n_{\text{tracks}}$ (4 inter-module gaps for each track) almost exactly. Drop rate reduces this linearly; ghost rate reduces it more weakly because ghost hits create candidate segments which the solver mostly rejects.

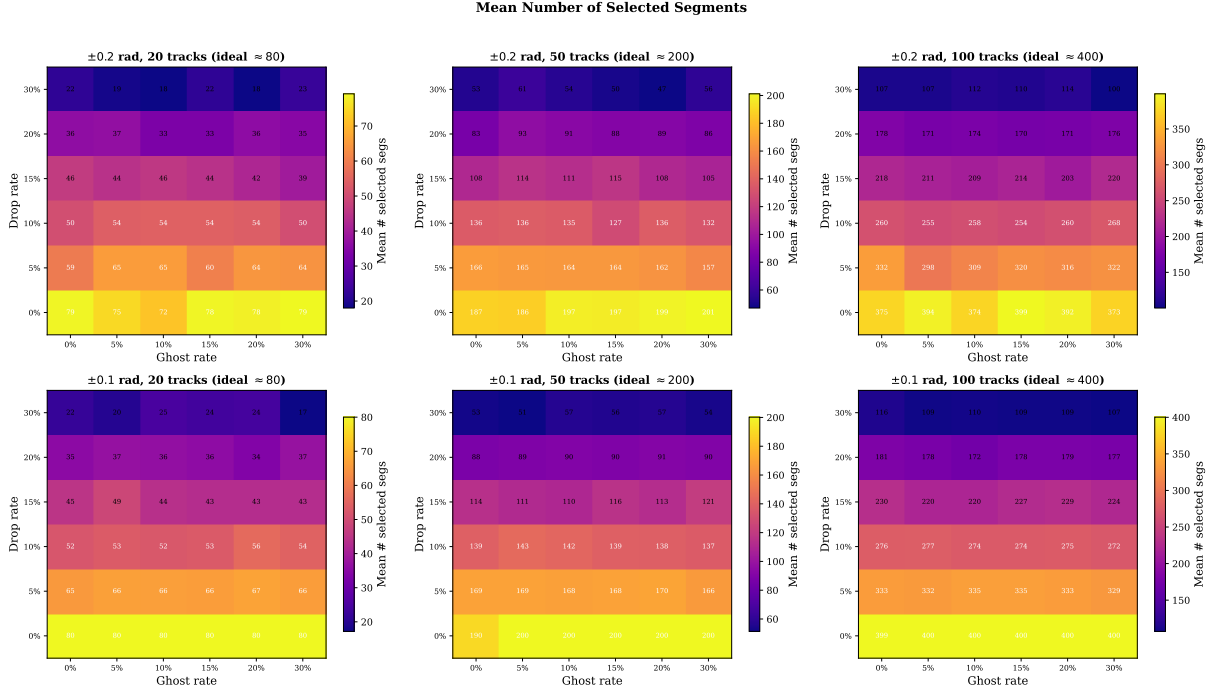


Figure 17: Mean number of selected segments per event across the noise grid, by angle and track multiplicity.

The true/false breakdown of the selected set (Fig. 18) confirms the solver’s discriminating power. False segments are essentially absent in clean configurations and remain a small minority even under heavy noise — typically $\lesssim 1\%$ of the selected set. One outlier appears at $(\phi = \pm 0.1, n = 50, \text{clean})$, where ~ 4.5 false segments per event survive with anomalously high activation ($\bar{x}_{\text{false}}^{\text{sel}} \approx 50$). This is consistent with an ill-conditioned A^{-1} in a nearly degenerate geometry; the affected fraction is small ($\sim 2\%$ of all selected) but flagged here for completeness.

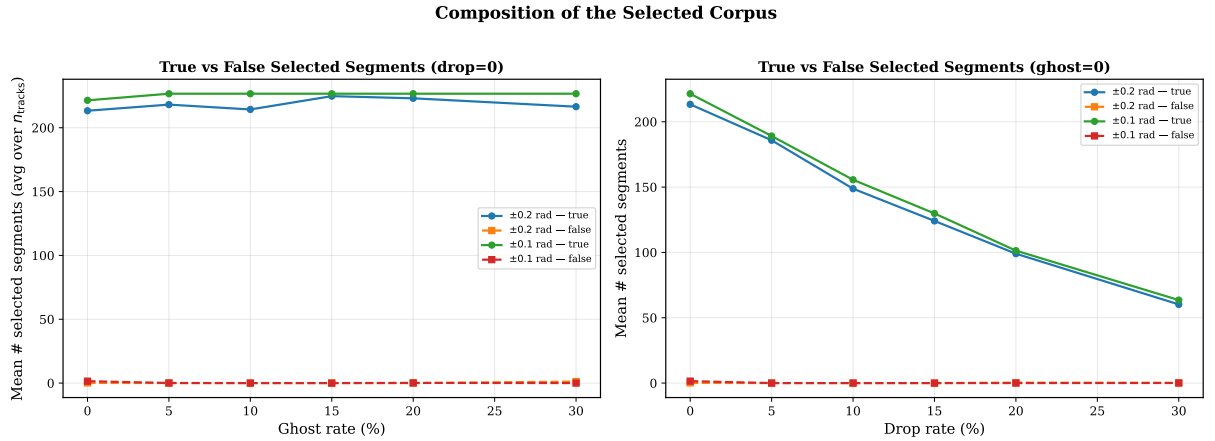


Figure 18: Mean number of true (filled) and false (hatched) segments in the selected corpus across noise configurations.

7.4 Selected-Segment Activation Spectrum

Within the selected corpus the activation distribution (Fig. 19) is sharply concentrated above the threshold for true segments, and the false-segment tail (when non-empty) sits at markedly higher values than the full-corpus baseline of $\delta/(\delta + \gamma) \approx 0.4$ — these are the borderline false segments the solver was unable to reject.

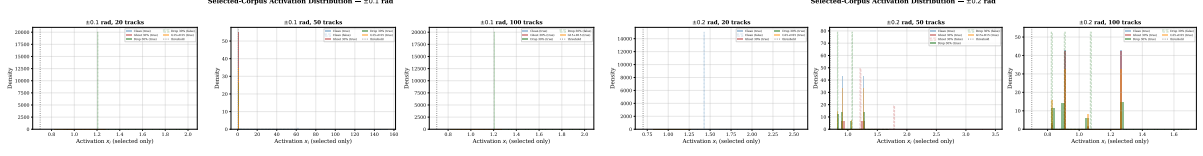


Figure 19: Activation x_i distribution within the selected corpus, split by truth label. Solid: true; stepped dashed: false. All densities lie above the threshold $T = 0.7$ by construction.

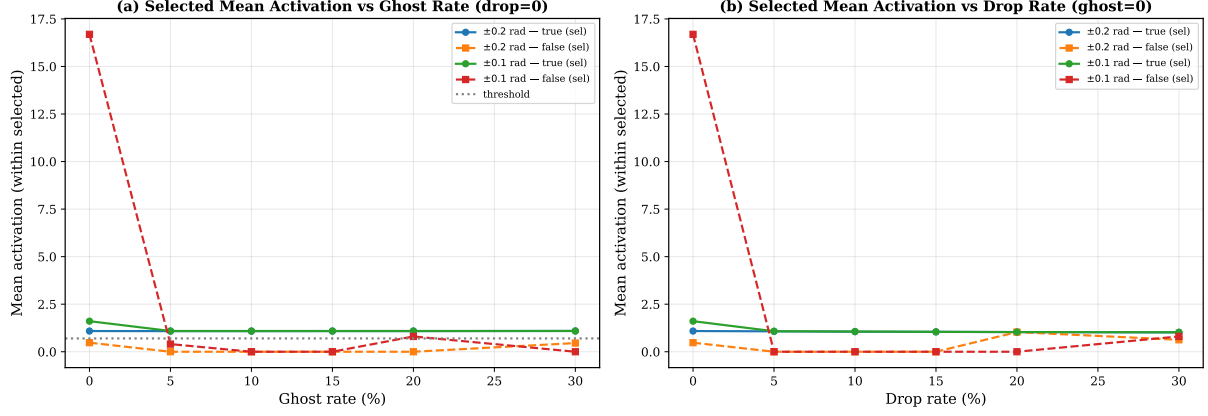


Figure 20: Mean true and false activations within the selected corpus vs ghost rate (left) and drop rate (right). False-segment activations are scarce and noisy due to the very small false population in the selected set.

7.5 Module-Level Hit Distribution

The per-module hit count in the selected corpus (Fig. 21) is uniform across modules in the clean baseline, as expected from the geometric prediction. The coefficient of variation across modules grows with drop rate (which preferentially removes hits at random) and is essentially flat in ghost rate (since ghost hits rarely survive selection).

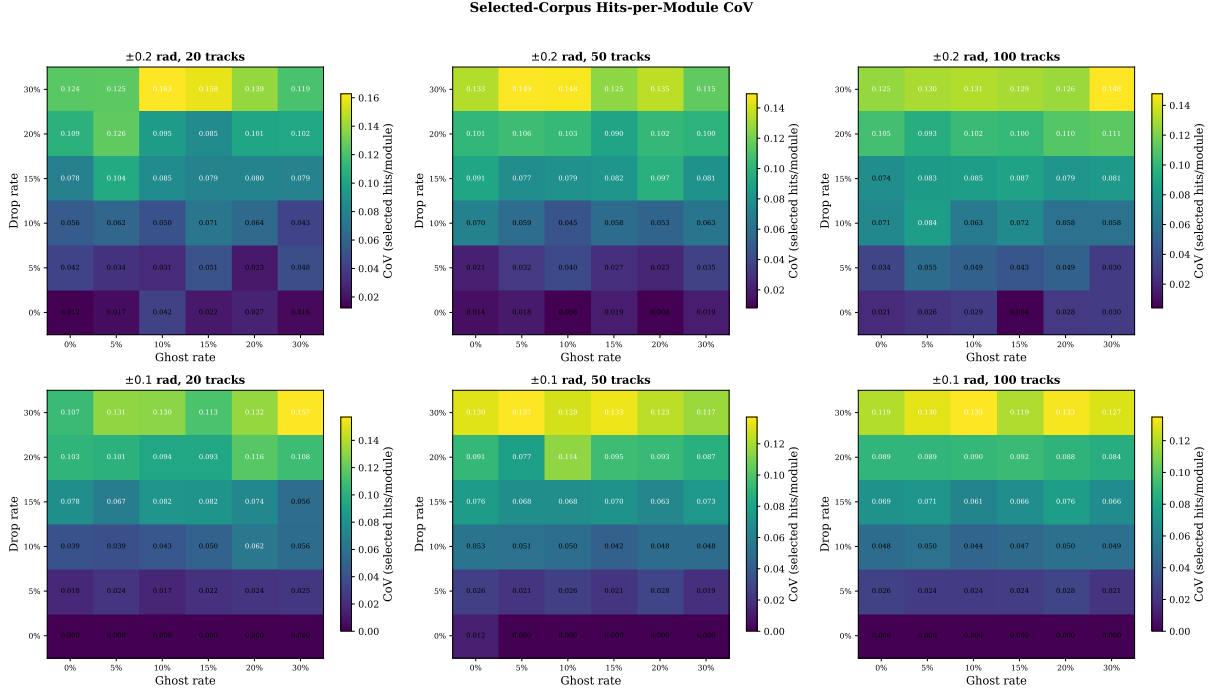


Figure 21: Coefficient of variation of selected hits across the five detector modules, vs ghost and drop rates.

7.6 Sensitivity of Selected-Corpus Spread

Figure 22 shows that the selected-corpus occupancy standard deviation is essentially constant at $\sigma \approx 0.49$ — exactly the binomial spread for occupancy values in $\{1, 2\}$ with $\sim 2/5$ probability of occupancy 1 — across the entire ghost and drop sweep. This is a strong robustness statement: the geometric $\{1, 2\}$ structure of the selected corpus is preserved across all noise regimes tested.

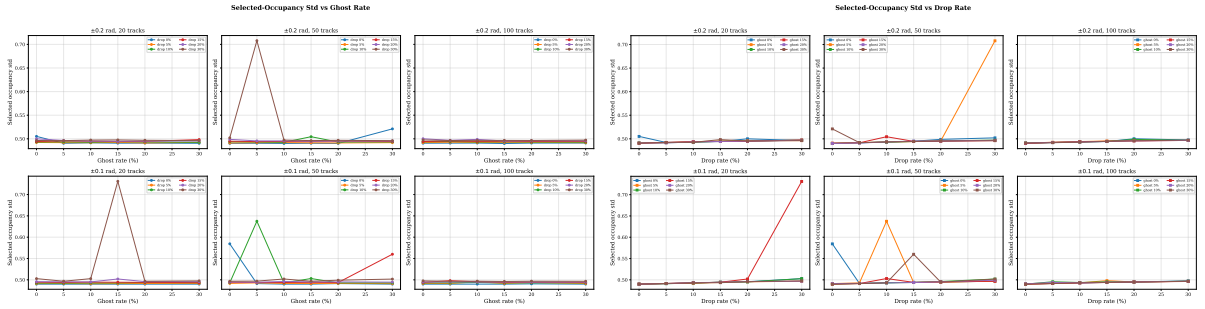


Figure 22: Standard deviation of selected-corpus occupancy vs ghost rate (left) and drop rate (right). Stable near 0.49.

Summary of the selected-corpus analysis. The classical solver, applied to the Hamiltonian-built corpus and threshold at $x_i > 0.7$, produces a segment set whose geometric properties are exactly what an idealised tracker would expect: $\text{occ} \in \{1, 2\}$ with mean 1.6, $4n_{\text{tracks}}$ selected segments, and a true/false ratio approaching the truth label to within $\sim 1\%$. The segment grass — anomalous-occupancy hits in the full corpus — does not propagate to the reconstruction stage.

8 Summary and Recommendations

The “segment grass” occupancy sub-peaks at $\pm 0.2\text{rad}$ are a **geometric clipping artefact**, not a feature of the Hamiltonian or the segment-competition mechanism. The systematic ghost/drop-rate sweep (Section 6) confirms this and reveals additional insights. The causal chain is:

1. The PV z -coordinate is drawn from $\mathcal{N}(0, 50^2)$ mm, frequently yielding $z_{\text{PV}} < -15$ mm.
2. Tracks at the maximum angle $\phi = \pm 0.2\text{rad}$ have slope $|t_x| \approx 0.203$, causing transverse displacements of up to $|t_x| \times (z_4 - z_{\text{PV}}) > 50$ mm at the last module.
3. These tracks miss the detector active area and produce no hit on the outer modules.
4. The reduced hit count on module k lowers the occupancy of hits on module $k-1$, creating sub-peaks in the histogram.
5. The effect is absent at $\pm 0.1\text{rad}$ because the corresponding z_{crit} values are $> 5\sigma$ from the PV mean.

Possible mitigations:

- Reduce the PV z spread (e.g. $\sigma_z = 10$ mm).
- Increase the detector half-width (e.g. $L_x = 100$ mm).
- Accept the effect as a realistic feature of finite detector acceptance and document it.

None of these sub-peaks affect the conclusions of the hit-competition study, because the segment-competition (γ -term) mechanism operates correctly on the segments that *are* constructed. The clipping merely reduces the total number of segments, which is a geometric acceptance effect independent of the Hamiltonian optimisation.

Additional findings from the noise sweep:

- Reconstruction efficiency is robust to ghost hits but degrades linearly with the drop rate ($100\% \rightarrow 30\%$ at 30% drop).
- The ghost and clone fractions in reconstructed tracks remain $\sim 0\%$ even at 30% ghost rate, confirming that the γ -term successfully suppresses false segments.
- Ghost hits inflate the Hamiltonian by up to 60% (at 100 tracks, 30% ghost), increasing computational cost without improving physics reach.
- True-segment activations degrade only with the drop rate (from 1.09 to 0.83 at 30% drop), while false-segment activations remain suppressed at the baseline ≈ 0.4 .

Findings on the selected (post-solver) corpus (Section 7):

- The mean selected occupancy is $\langle \text{occ}_{\text{sel}} \rangle \approx 1.6$ with $\sigma \approx 0.49$, matching the geometric prediction $(2 \cdot 1 + 3 \cdot 2)/5 = 1.6$ for a 5-module detector.
- Anomalous-occupancy fraction collapses from a median of 0.98 in the full corpus to 0.0 in the selected corpus across the entire 216-point sweep. The classical solver acts as a near-perfect grass filter.
- Selected segments per event scale as $4n_{\text{tracks}}$ in clean runs, decreasing approximately linearly with drop rate.

- False segments form $\lesssim 1\%$ of the selected set under all noise conditions tested, with one minor outlier at $(\phi = \pm 0.1, n = 50, \text{clean})$ flagged for follow-up.
- The selected-corpus structure is robust to both ghost and drop noise across the tested ranges, supporting the use of the $x_i > 0.7$ threshold as a stable reconstruction primitive.